Wednesday Reading Assessment: Unit 0, Review of 135A

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1 Memory Bank

- $\vec{a} = a_x\hat{i} + a_y\hat{j}$ ... Component notation for 2D vector
- $\vec{a} + \vec{b} = (a_x + b_x)\hat{i} + (a_y + b_y)\hat{j}$ ... Adding vectors
- $\vec{a} - \vec{b} = (a_x - b_x)\hat{i} + (a_y - b_y)\hat{j}$ ... Subtracting vectors
- $|\vec{a}| = \sqrt{a_x^2 + a_y^2}$ .. Magnitude of a 2D vector
- $a_x = |\vec{a}|\cos\theta$ ... x-component of a 2D vector
- $a_y = |\vec{a}|\sin\theta$ ... y-component of a 2D vector
- **Couloumb's Law**: Let two charges with magnitudes q_1 and q_2 be separated by a displacement $\vec{r}$. The force between them is
$$\vec{F} = \frac{kq_1q_2}{r^2}\hat{r} \tag{1}$$
The vector $\hat{r}$ is a unit vector in the direction of $\vec{r}$, and r is the magnitude. The constant k is $k = 8.988 \times 10^9$ N m^2 C^{-1}.
- $\vec{F} = q\vec{E}$ ... The electric field $\vec{E}$ causes the Coulomb force on a charge q.
- **Couloumb Field**: Let a charge with magnitude q be separated by a displacement $\vec{r}$ from a *test charge*. The E-field generated by q is
$$\vec{E} = \frac{kq}{r^2}\hat{r} \tag{2}$$
The vector $\hat{r}$ is a unit vector in the direction of $\vec{r}$, and r is the magnitude. The constant k is $k = 8.988 \times 10^9$ N m^2 C^{-1}.

2 Warm-Up Exercises

1. Perform the following unit conversions:
   - Convert 320 gm to kg:
   - Convert 2.4×10^5 cm s^{-1} to km hr^{-1}:
   - Convert the density of ice, 0.9167 gm cm^{-3}, to kg m^{-3}.

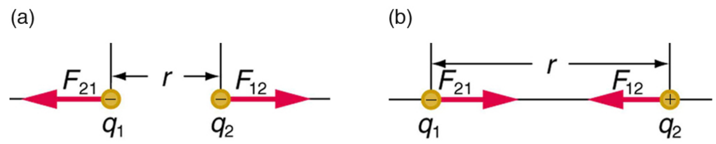

Figure 1: Coulomb forces between charges.

2. Let $\vec{x} = 0.5\hat{i} - 0.5\hat{j}$, and $\vec{y} = -0.5\hat{i} + 0.5\hat{j}$. (a) Calculate $\vec{x} + \vec{y}$. (b) Calculate $\vec{x} - \vec{y}$. (c) What is the magnitude of $\vec{a}$? (d) What is the magnitude of $\vec{b}$?

3. Let $\vec{F}$ be a 10 N force that makes a 60 degree angle with the x-axis. (a) What is the x-component of this force? (b) What is the y-component? (c) Suppose this force displaces a system 3 m along the x-axis. What is the work done? *Recall that* $W = \vec{F} \cdot \Delta\vec{x}$.

4. Suppose a charge $q_1 = -1$ nC is located 1 cm away from a charge $q_2 = +1$ nC. What is the magnitude and direction of the force between the charges? To interpret the direction, consider Fig. 1.

5. Suppose four equal charges q form a square with side length a. What is the magnitude of the electric field at the center of the square?